

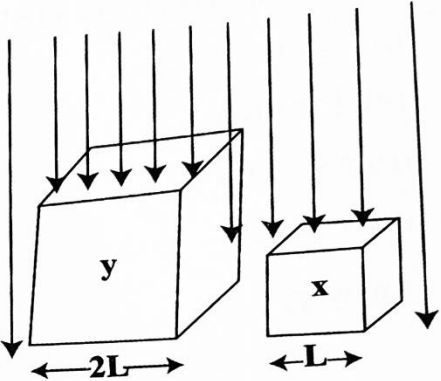
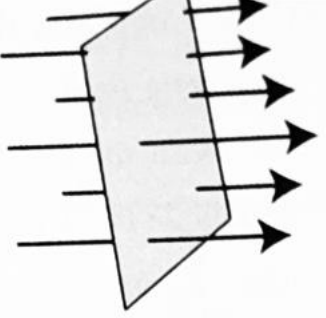
Lesson (1): Meaning of Magnetic Field

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	The area around a magnet (or a current – carrying a conductor) where magnetic forces can act.	
2	It is the total number of magnetic flux lines passing through a surface. (الدور الثاني 2023)	
3	It is the total number of magnetic flux lines passing normally through a unit area around the point.	
4	It is the magnetic flux density when the total number of magnetic flux lines passing normally through a unit area around the point is 1 Wb	
5	It is the portion of space near a magnetic body or a current-carrying body in which the magnetic forces due to the body or current can be detected.	

2. Put circle around the correct answer

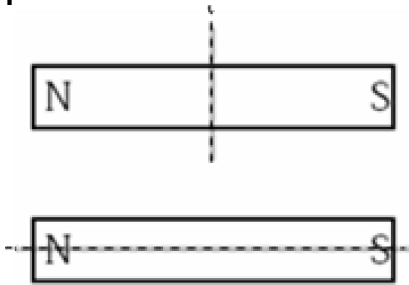
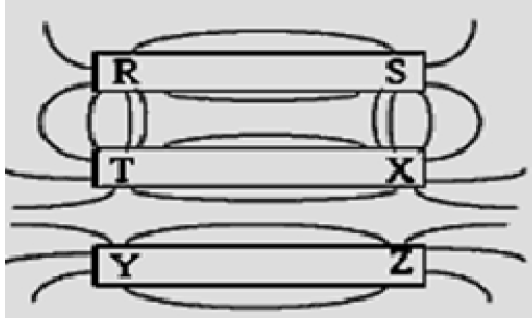
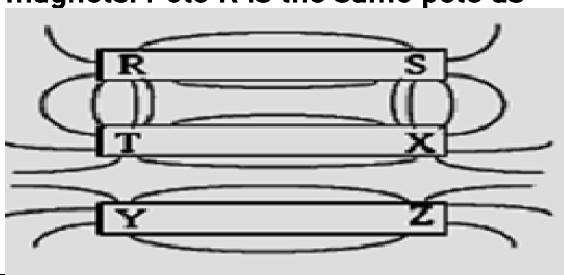
No.	Question	Choices	Answer
1	Magnetic lines can be directed from south to north	a) Outside the magnet b) Inside the magnet c) Near the center d) Anywhere	
2	What is the origin of magnetism	a) vibration motion of atoms of iron b) motion of the electrons in an orbit around the nucleus c) The spin of the electrons around its axis d) motion of the electrons in an orbit around the nucleus and the spin of the electrons around its axis	
3	To describe the magnetic field, we need to know	a) the shape of magnetic field only b) The direction of the magnetic field only c) The magnetic field density only d) all of the above	
4	The total number of magnetic flux lines passing through a surface is	a) magnetic flux b) magnetic flux density c) electric current d) electric current intensity	

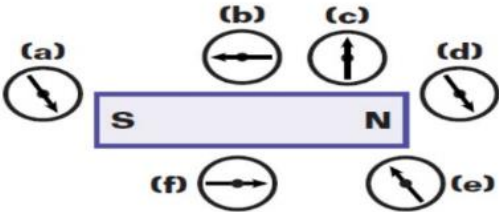
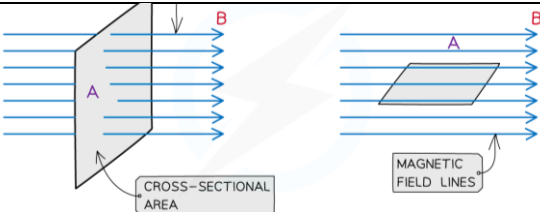
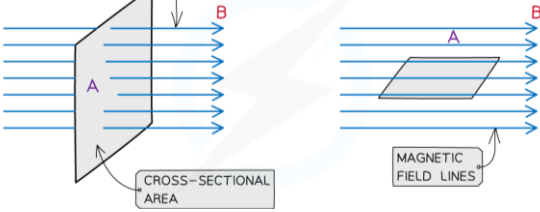
5	The magnetic field density	a) number of magnetic lines per 1 meter squared b) number of magnetic lines per 1 centimeter squared c) number of magnetic lines per 4 meters squared d) number of magnetic lines per 4 centimeters squared	
6	Theta at the equation ($\Phi_m = B A \cos\theta$)	a) angle between magnetic line and surface b) angle between magnetic flux and surface c) angle between normal to the surface and magnetic lines	
7	Two cubes (x) and (y) that have side lengths (L) and (2L) respectively are affected by a uniform magnetic field perpendicularly on the upper surface of each of them as shown in the opposite figure, so which of the following relations represents the relation between the magnetic flux that passes through the upper surface of them. 	a) $(\Phi_m)_y = \frac{1}{4}(\Phi_m)_x$ b) $(\Phi_m)_y = (\Phi_m)_x$ c) $(\Phi_m)_y = 2(\Phi_m)_x$ d) $(\Phi_m)_y = 4(\Phi_m)_x$	
8	 In which of the following cases, the flux which passes through the coil becomes $\left(\frac{\Phi_m}{2}\right)$ if the flux at the figure equal Φ_m		

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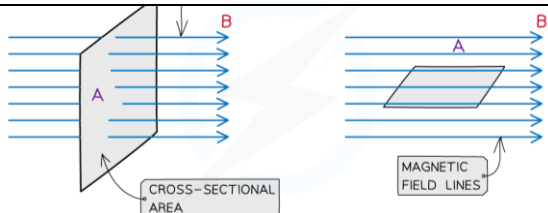
	(A) (B) (C) (D)	
9	The opposite diagram shows a rectangular coil that is moving with constant velocity towards the right direction of the page penetrating a uniform magnetic field which is normal into the page, so the relation between the magnetic flux (ϕ_m) that passes through the coil during its motion from position (A) to position (B) and the time (t) is ... (A) (B) (C) (D)	
10	A square of area of cross section 10^{-2} m^2 is placed with its axis perpendicular to a uniform magnetic field of flux density 1 T. Then the total magnetic flux through the coil is ... (الدور الأول 2024)	a) 10^{-2} Wb b) $5 \cdot 10^{-3} \text{ Wb}$ c) $5\sqrt{3} \cdot 10^{-3} \text{ Wb}$ d) $5 \cdot 10^{-2} \text{ Wb}$
11	When the density of magnetic field in a given area is increased to its double value. The magnetic flux through this area would be... (الدور الأول 2023)	a) Unchanged b) Doubled c) Increased four times d) Increased eight times
12	A rectangular wire loop of its dimensions (5cm * 8 cm) is at rest in a uniform magnetic field of flux density 2T that is directed into the page. The plane of the loop is perpendicular to the field, the total magnetic flux through the loop is (تجريبى)	a) Zero b) 0.002 Wb c) 0.0052Wb d) 0.008 Wb

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13	Magnetic lines can be directed from south to north	a) Outside the magnet b) Inside the magnet c) Near the magnet	
14	A magnet usually has magnetic pole(s).	a) One b) Two c) Three d) Four	
15	Two bar magnets are to be cut in half along the dotted lines shown. None of the pieces are rotated. After the cut: 	a) The two halves of each magnet will attract each other b) The two halves of each magnet will repel each other c) The two halves of the top magnet will repel, the two halves of the bottom magnet will attract d) The two halves of the top magnet will attract, the two halves of the bottom magnet will repel	
16	The diagram to the right depicts iron filings sprinkled around three permanent magnets. Pole S is the same pole as 	a) T and Y b) T and Z c) X and Y d) X and Z	
17	The diagram to the right depicts iron filings sprinkled around three permanent magnets. Pole R is the same pole as 	a) T and Y b) T and Z c) X and Y d) X and Z	
18	Which of the compass-needle orientations in the figure might correctly describe the magnet's field at that point?	a) a and b b) b and c c) c and d d) f and e	

			
19	Electron's motion in an atom is	a) The motion of the electrons in an orbit around the nucleus b) The spin of the electrons around its axis c) Both a and b d) Electrons don't have any motion.	
20	At a maximum magnetic flux, the magnetic flux equals to:	a) BA b) Zero c) One d) None of the above	
21	At a minimum magnetic flux, the magnetic flux equals to	a) BA b) Zero c) One d) None of the above	
22	At a maximum magnetic flux, the cosine equals to	a) BA b) Zero c) One d) None of the above	
23	At a minimum magnetic flux, cosine equals to:	a) BA b) Zero c) One d) None of the above	
24	 From the figure above, which one has a maximum magnetic flux?	a) Figure on the left b) Figure on the right c) Magnetic flux is constant regardless. d) An inclined surface	
25	 From the figure above, which one has a minimum magnetic flux?	a) Figure on the left b) Figure on the right c) Magnetic flux is constant regardless. d) An inclined surface	

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26	 From the figure above, which one has a maximum magnetic flux density?	a) Figure on the left b) Figure on the right c) Magnetic flux density is constant regardless. d) An inclined surface	
27	When the density of magnetic field in a given area is increased to its double value. The magnetic flux through this area would be ...	a) Unchanged b) Doubled c) increased four times d) increased eight time	

3. Correct the underlined in the following statement

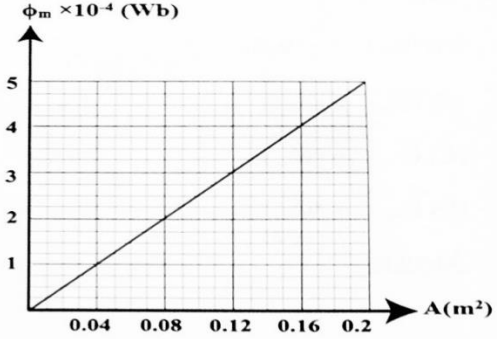
No.	Question	Answer
1	The maximum value of magnetic flux cutting a certain area when the magnetic flux lines is <u>parallel</u> to that area. (الدور الأول 2023)	
2	Iron filings can indicate the <u>direction</u> of magnetic fields	
3	Magnetic flux the total number of magnetic flux lines passing through a <u>unit area</u> .	

4. True or False

No.	Question	Answer
1	Magnetic field lines can cross each other	
2	Magnetic field lines inside the magnet directed from south to north	
3	A compass is used to indicate the direction of a magnetic field.	
4	The motion of the electrons in an orbit around the nucleus is the source of magnetism.	
5	The magnetic field lines are imaginary lines used to represent magnetic fields	
6	Magnetic field lines are directed from south magnetic pole to north magnetic pole.	
7	Magnetic field lines are not allowed to cross each other	
8	Iron filings are used to show the shape of magnetic fields	
9	Iron filings can indicate the direction of magnetic fields	

10	Magnetic field is produced around a conductor when an electric current passing through it.	
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5. Answer the following question

No.	Question	Answer
1	Many rectangular coils of different areas are placed one at a time in a uniform magnetic field such that each of them is inclined to the normal of the field by an angle of 60°. The opposite graph represents the relation between the total flux (Φ_m) that passes through the coil and the area of the coil (A), so the magnetic flux density affects the coil is... 	
2	When a square coil of area 0.2m^2 putted normal to magnetic flux density (B) the magnetic flux passes through it is Φ_m , what is the angle between the coil and magnetic flux density to make the magnetic flux passes through it becomes $\Phi/2$	
3	Rectangular coil of cross-sectional area (a) putted perpendicular to magnetic lines of magnetic flux density (B) so the magnetic flux passes through it become 10wb, if the magnetic flux increases by 2.5T the magnetic flux passes through it becomes 50wb so (b) =	
4	At the opposite figure, If the area of coil (x) is double the coil (y), so the ratio between the magnetic flux which penetrate coil (x) to that penetrate coil (Y)	

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5	An aluminum window frame has a width of 40 cm and length of 73 cm. The frame is hinged along the vertical edge AC. When the window is closed, the frame is normal to the Earth's magnetic field with magnetic flux density $1.8 \times 10^{-5} \text{ T}$ a) Calculate the magnetic flux through the window when it is closed b) Sketch the graph of the magnetic flux against angle between the field lines and the normal when the window is opened and rotated by 180°	
6	A rectangular loop has dimensions of 0.50m and 0.60m. The values of magnetic field is 0.02T with an angle of 45°. Calculate the magnetic flux at the surface.	
7	A perpendicular field of force of magnitude 0.1 T is applied to an oblong loop with dimensions of three cm by 5 cm. How does one calculate the magnetic flux through the loop?	

Lesson (2): Magnetic field due to current passing through a straight conductor

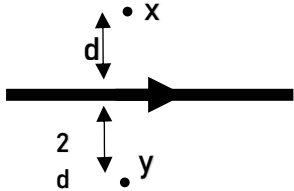
1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	Magnetic field is produced around a conductor when an electric current passing through it	
2	It is the ability of medium to penetrate the magnetic flux lines	
3	If the thumb points in the direction of the current, the rest of the fingers around the wire will indicate the direction of the magnetic field due to the current.	

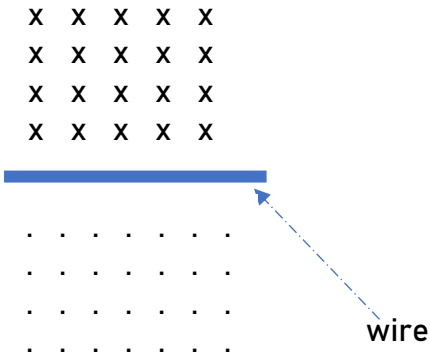
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	magnetic field due to a current in straight wire	
2	direction of magnetic field due to a current in straight wire	

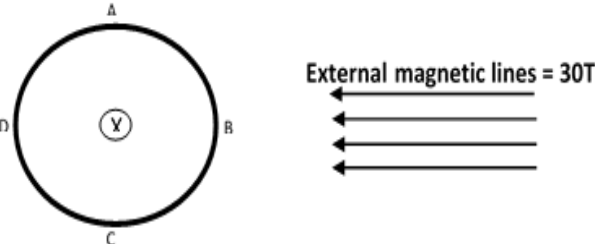
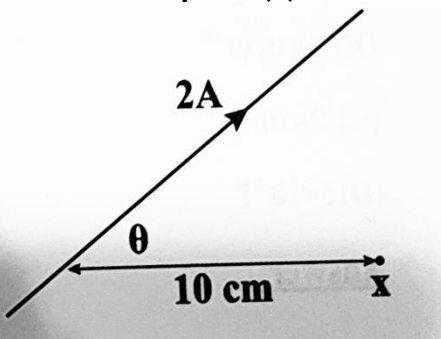
3. Put circle around the correct answer

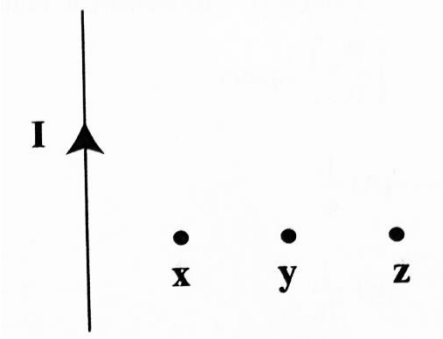
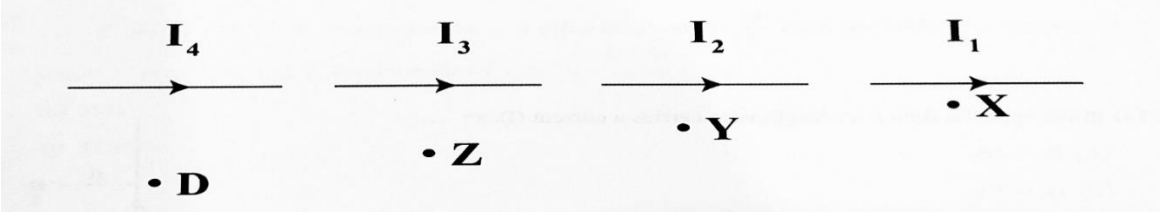
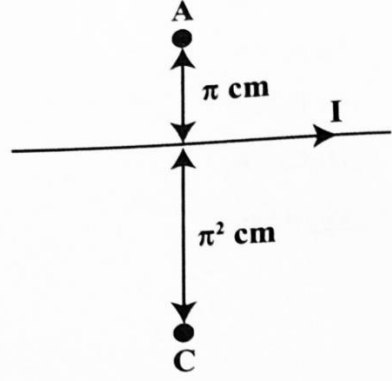
No.	Question	Choices	Answer
1	The ratio between the magnetic flux density at point (x) to that at point (y) $\frac{B(x)}{B(y)}$ 	a) 1/1 b) 1/2 c) 2/1 d) 1/4	
2	To identify the direction of the magnetic field lines around a straight conductor carrying-current we use	a) Ampere's right-hand rule b) Ampere's left-hand rule c) Fleming right-hand rule d) Fleming left-hand rule	
3	A long straight wire conductor is placed below a compass as shown in the top view figure. When a large current flows in the conductor as shown, the N pole of the compass:	a) Points to the north b) Points to the south c) Points to the east d) Points to the west	

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4	Permeability measured by	a) T.m/A b) Ampere c) Wb/m ² d) Weber	
5	a straight wire carrying a current of intensity 4 A if the magnetic flux density at a point at a certain distance from its axis 0.00002 Tesla. the distance of the point from the axis of the wire	a) 0.04 m b) 0.16 m c) 0.004 m d) 0.016 m	
6	The magnetic flux density at a distance d from the long straight current carrying wire is 0.4T the magnetic flux density at a distance 2d is	a) 0.1T b) 0.2T c) 0.8T d) 1.6T	
7	At double the distance from a long current-carrying wire, the strength of the magnetic field produced by that wire will be	a) doubled b) increased four times c) decreased to 1/2 of its original value d) decreased to 1/4 of its original value	
8	Rule used to determine the direction of Magnetic field due to a current in a straight wire	a) Ampere right hand rule b) Ampere left hand rule c) Fleming right hand rule d) Fleming left hand rule	
9	Show the direction of current in the straight wire in each of the following drawing: x 	a) upward b) downward c) leftward d) rightward	
10	The shape of magnetic lines due to a straight wire carrying electric current is:	a) Aligned in concentric circles around the wire.	

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		b) Farther apart from each other near the wire. c) Closer together as distance from the wire increases. d) Aligned in concentric triangles around the wire.	
11	The magnetic flux density produced around a wire carrying current increases by (as the other factor are constant)	a) getting away from the wire. b) increasing the length of the wire. c) increasing the current intensity flowing through the wire. d) decreasing the current intensity flowing through the wire.	
12	There is a straight wire carrying current to produce magnetic flux density of 30 T on the circumference of the circle ABCD putted in an external magnetic flux of 30T Which point has a resultant magnetic flux of zero T. 	a) Point A b) Point B c) Point C d) Point D	
13	The magnetic flux density at a distance of 10cm in air from the center axis of a long wire carrying a current of 10A equals	a) $2 \times 10^{(5)}\text{T}$ b) $2 \times 10^{(-5)}\text{T}$ c) $2 \times 10^{(-7)}\text{T}$ d) $2 \times 10^{(-9)}\text{T}$	
14	In the opposite figure, the magnetic flux density due to the flow of the electric current in the wire at point (x) is 	a) equal to $4 \times 10^{-6}\text{T}$ b) greater than $4 \times 10^{-6}\text{T}$ c) smaller than $4 \times 10^{-6}\text{T}$ d) indeterminable	

15	A long straight wire carrying electric current of intensity (I) as in the figure. Which one from the following expressions represents correctly the magnetic flux density (B) that is produced at points (x), (y), (z) due to the passing current through the wire? 	a) $B_x > B_y$ b) $B_y < B_z$ c) $B_z > B_x$ d) $B_y = B_z$
15	The opposite figures shows four straight carrying different currents (I_1, I_2, I_3, I_4). If the magnetic flux densities at points (X, Y, Z, D) are equal. So, the greatest electric current intensity is  a) I_1 b) I_2 c) I_3 d) I_4	
16	The opposite figure represents, a straight wire carrying current (I). Two points (A), (C) beside the wire, so its magnetic flux density at point (A) is (B_A) and its magnetic flux density at point (C) is (B_C), so the ratio $\left(\frac{B_A}{B_C}\right)$ equals (A) $\frac{1}{\pi}$ (B) $\frac{1}{2\pi}$ (C) 2π (D) π 	

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17	25) Which of the following figures represents correctly the generated magnetic field due to the flow of a direct electric current in a straight wire? (A) (B) (C) (D)	
18	26) A wire carrying current passes through a uniform magnetic field, as shown. At which point is the magnetic field the strongest? (A) (A) (B) (B) (C) (C) (D) (D)	
19	Which the graph correctly gives the relation between the magnitude of the magnetic field density (B) outside an infinitely long straight current-carrying wire and the distance of point from the wire (d)? (الدور الأول 2023) (a) (b) (c) (d)	
20	Two metallic rings A and B, If a ring A of radius r, current I is flowing and in ring B of radius $2r$, current $2I$ is flowing, then (الدور الأول 2024) $\frac{\text{the density of magnetic field at center of ring A}(B_A)}{\text{the density of magnetic field at center of ring B}(B_B)}$ a) $\frac{1}{4}$ b) $\frac{1}{2}$ c) $\frac{1}{1}$ d) $\frac{4}{1}$	
21	If the magnitude of the magnetic field a distance d from a very long, straight, current-carrying wire is B, at what distance from the wire will the field have magnitude $4B$? (تجريبی) a) $8d$ b) $4d$ c) $d/4$ d) $d/8$	

4. Correct the underlined in the following statement

No.	Question	Answer
1	The magnetic field density generated from a <u>solenoid</u> carrying-current is inversely proportional to the distance to that point.	

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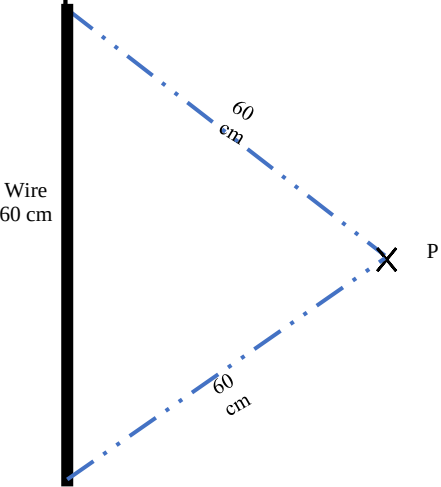
5. True or False

No.	Question	Answer
1	Magnetic field is produced around a conductor when an electric current passing through it	
2	When an electric current flows in a straight wire it generates a magnetic field surrounding the straight wire.	
3	To determine the direction of the magnetic field resulting from an electric current in a wire, by using Ampere Right Hand Rule.	

6. Answer the following question

No.	Question	Answer
1	Draw the shape of magnetic field due to a current in a straight wire	
2	Mention the rule or method used to determine the direction of the magnetic field resulting from an electric current in a straight wire (تجريبی)	
3	An electric current of intensity 10 A is flowing in a long straight wire. Calculate the density of magnetic field at a distance 0.1m from the wire. (Permeability of air = $4\pi \times 10^{-7} \text{ Wb /A .m}$) (الدور الأول 2024)	
4	What is the current passes in a wire of length 26.4 cm which curved in the form of an arc of a circle of radius 5.6 cm to form magnetic flux density of $8.25 \times 10^{-4} \text{ T}$ at its center.	
5	Determine the magnetic flux density at a distance of 7 cm in air from the center of a long wire carrying a current 2.5 A. (Permeability of air = $4\pi \times 10^{-7} \text{ Wb /A .m}$)	
6	A long wire carries a current of 1 mA. Find the magnitude of the magnetic field 2 m away from the wire.	

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	(Permeability of air = $4\pi \times 10^{-7} \text{ Wb /A .m}$)	
7	Find the current in a long straight wire that would produce a magnetic field twice the strength of the Earth's at a distance of 5.0 cm from the wire. (The Earth's field is about $5.0 \times 10^{-5} \text{ T}$)	
8	A long wire of length 60 cm with a 60mA passing through it. Calculate the magnetic flux density at point P shown in the picture below. 	

7. Compare between each of the following

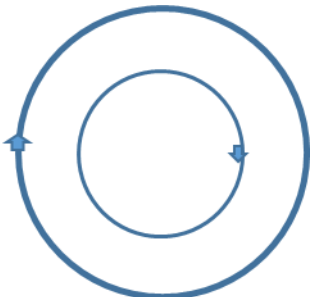
No.	Question		Answer
1	Compare between the application of the following		
	Point of comparison	ampere's circular law	ampere's right-hand rule
	Application		

Lesson (3): Magnetic field due to current passing through a circular coil (loop)

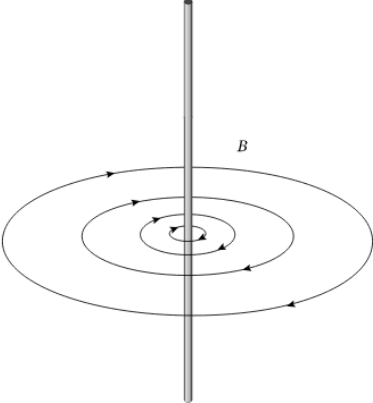
1. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	magnetic field due to a current in circular coil	
2	Direction of magnetic field due to a current in circular coil	

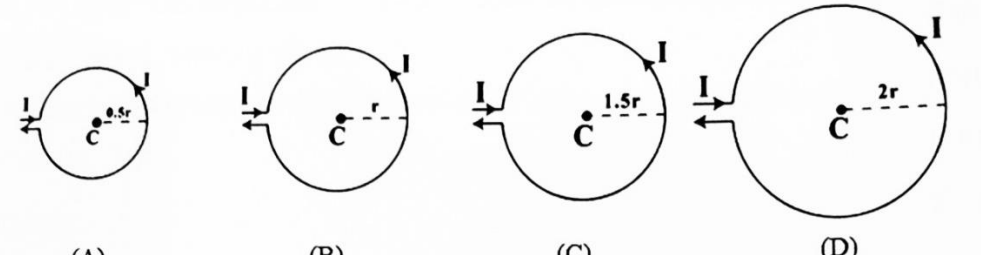
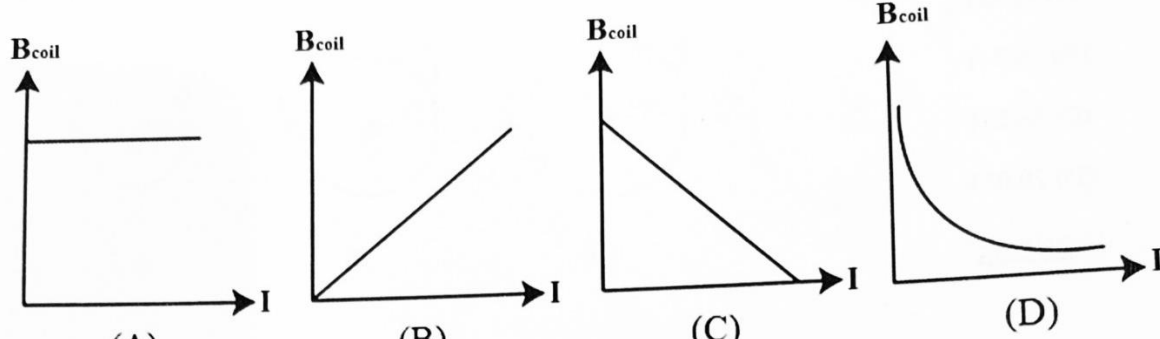
2. Put circle around the correct answer

No.	Question	Choices	Answer
1	Two concentric metal rings in one plane, each carries a current of intensity (I) as shown in figure, the direction of the magnetic flux at the common center (m) is: 	a) Rightwards b) Leftwards c) Into the page d) out of the page	
2	Produce a magnetic field looks like a short bar magnet	a) Solenoid coil b) Circular coil c) Straight wire d) All the previous	
3	The magnetic field due to a current in a circular coil has	a) Uniform magnetic field everywhere b) uniform magnetic field at the coil center c) Uniform magnetic field away from the coil d) no uniform magnetic field	

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4	The circular magnetic flux lines are closer together _____ the wire and _____ apart from each other as the distance from the wire increases.	a) Near, farther b) Farther, farther c) Near, near d) Farther, near	
5	We know the magnetic field lines' direction due to a circular loop using	a) Right Corkscrew rule b) Maxwell's Screw Rule c) Clockwise rule d) All the above	
6	 From the figure above, what is the direction of the current?	a) Upwards b) Downwards c) Both a and b d) None	
7	We can get the magnitude of magnetic field at the center of circular coil because a) Lines are non-uniform at the center b) lines are uniform at the center c) Lines directed from north to south at the center d) lines directed from south to north at the center		
8	The magnetic field passing through the center of a circular ring carrying current is: (A) Circular. (B) Ellipse. (C) Straight line. (D) Zero.		
9	The magnetic flux density at the center of a circular coil of radius 11cm carrying a current 1.4 Ampere. If the coil consists of 20 turns equals: (A) $9.8 \times 10^{-9} \text{T}$ (B) $3.2 \times 10^{-4} \text{T}$ (C) $1.6 \times 10^{-4} \text{T}$ (D) $7.7 \times 10^{-6} \text{T}$		

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10	Four metallic circular rings of different radii and same current intensity pass in each of them as in the figure, so the smallest magnetic flux density is generated at the center of the ring  (A) (B) (C) (D)	
11	Which graph of the following represents the relation between the magnetic flux density of coil (B_{coil}) and electric current intensity (I) passing through the coil? (When the other factors are constant)  (A) (B) (C) (D)	
12	1) Wire of a certain length carries a steady current. It is first bent to form a circular coil of one turn. The same wire is next bent to form a circular coil of four turns. The ratio of the first magnetic flux density to the second is: (A) $1/16$ (B) $16/1$ (C) $4/1$ (D) $1/4$	
13	To obtain small magnetic field from solenoid carrying current we have to (When other factors are constant). (A) decrease the number of turns of the solenoid. (B) decrease the length of the solenoid. (C) increase the current passing through it. (D) increase the number of turns of the solenoid.	

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14	The magnetic flux density increases at the center of a circular coil carrying an electric current by decreasing a) cross sectional area of the coil b) the number of the coil turns c) the current intensity in the coil d) the permeability of the coil core	
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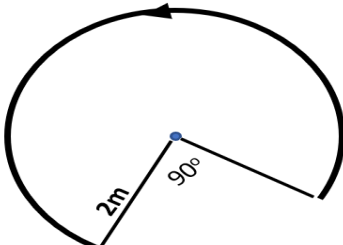
3. Correct the underlined in the following statement

No.	Question	Answer
1	The magnetic flux density at center of a circular coil carrying a current decrease by <u>decreasing</u> the cross-sectional area of the coil. (الدور الأول 2023)	

4. Answer the following question

No.	Question	Answer
1	A square of side 20cm putted normal to magnetic flux Density of 2T, if it is reformed to be a circular coil and also placed normal to the previous magnetic field so the magnetic flux passing through the circular coil equals to	
2	Determine the magnetic flux density at the center of a circular loop of radius 11 cm carrying a current of 1.4 A. if the wire loop consists of 20 turns. (Permeability of air = $4\pi \times 10^{-7}$ Wb /A .m)	
3	There is a straight wire of length 360 m that has been wrapped around in a circle 7 times. Find its magnetic flux density given that the current is 1A.	
4	A current of intensity 0.1A passes through a circular coil of diameter 12.56 cm and number of turns 100 turns. Calculate the magnetic flux density at the center of coil	
5	what is the shape of the shape magnetic field due to a current in a circular coil.	

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6	Mention the rule or method used to determine the direction of the magnetic field resulting from an electric current in a circular coil. (تجريبی)	
7	A current passes in a wire of length 26.4cm curved in the form of an arc of a circle of radius 5.6cm to form magnetic flux density of $8.25 \times 10^{-6} \text{ T}$ at its center. Find the current intensity in the wire.	
8	A circular loop of radius 0.1m carries a current of 10A, what is the magnetic flux density at its center?	
9	What is the current intensity in a ring of radius 10cm such that the magnetic flux density at its center is $5 \times 10^{-5} \text{ T}$	
10	Calculate the intensity of electric current which passes through a circular coil of diameter 22 cm, 49 turns and produces a magnetic flux density in its center equal to $7 \times 10^{-5} \text{ T}$. (الدور الثاني 2023)	
11	The segment of wire in the figure carries a current of $I=5\text{A}$. Where the radius of circular arc is $R= 3\text{cm}$. Determine the magnitude and direction of the magnetic field at the origin. 	
12	the wire shown in the figure carries a current of 40A, find the magnetic induction at point (p) and its direction. 	

Unit 9

5. Compare between each of the following

No.	Question		Answer
1	Compare between the application of the following		
	Point of comparison	Maxwell's Screw Rule	polarity coil
	Application		

Lesson (4): Magnetic field due to current passing through a solenoid

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	It is the point at which the total magnetic flux density vanishes. (الدور الثاني 2023)	

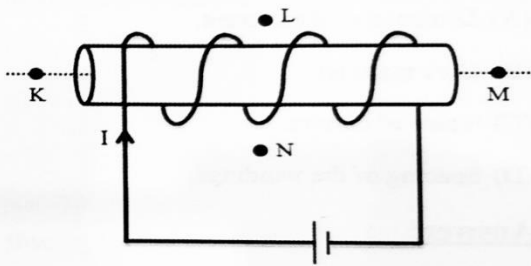
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	The density of magnetic field at a point along interior axis of a solenoid carrying an electric current. (الدور الأول 2024) (الدور الأول 2023)	
2	direction of magnetic field due to a current in solenoid	

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	Produce a magnetic field looks like a long bar magnet	a) Solenoid coil b) Circular coil c) Straight wire d) All the previous	
2	The magnetic field produces are directly proportional to the passes electric current on it	a) Solenoid coil b) Circular coil c) Straight wire d) All the previous	
3	Magnetic lines come from..... looks like long bar magnet	a) Solenoid coil b) Circular coil c) Straight wire d) All the previous	
4	Rule to determine the direction of the magnetic field resulting from an electric current in a solenoid	a) Fleming right hand rule b) Fleming left hand rule c) Polarity d) Ampere right hand rule	
5	(n) in the rule ($B = \mu n I$) for solenoid represented	a) Number of turns b) Number of turns per unit length c) Electric current d) Permeability	
6	The magnetic field in the core of a solenoid carrying-current is	a) that is of maximum value acts on the wire	

Unit 9

		b) that is of zero value acts on the wire c) will push the wire away from the wire d) will pull the wire towards the wire																
7	The magnetic field in the core of a solenoid carrying-current is	a) uniform outside the solenoid b) uniform along the solenoid axis c) directed from north pole to south pole d) non-uniform along the solenoid axis																
8	A solenoid is connected to a battery in a closed circuit, if its turns are compressed regularly. The magnetic flux density at a point on its axis inside it will (الدور الأول 2023)	a) increase b) decrease c) remain the same d) vanish																
9	The magnitude of magnetic flux density along the axis of solenoid is independent of: (A) Direction of the current. (B) Core material. (C) Value of current. (D) Spacing of the windings.																	
10	Which point that representing the north pole when a current passing through the solenoid? (A) K (B) L (C) N (D) M Answer:																	
11	A solenoid of length 20cm and number of turns 200 turns, carries electric current 0.5A. The magnetic flux density at the middle of the axis is (knowing that $\mu_{\text{iron}} = 2 \times 10^{-3} \text{T.m/A}$ and $\pi = 3.14$) <table><tr><td></td><td>If the medium is air</td><td>If the medium is iron</td></tr><tr><td>(A)</td><td>$6.28 \times 10^{-4} \text{T}$</td><td>$6.28 \times 10^{-4} \text{T}$</td></tr><tr><td>(B)</td><td>1T</td><td>$6.28 \times 10^{-4} \text{T}$</td></tr><tr><td>(C)</td><td>$6.28 \times 10^{-4} \text{T}$</td><td>1T</td></tr><tr><td>(D)</td><td>1T</td><td>1T</td></tr></table>				If the medium is air	If the medium is iron	(A)	$6.28 \times 10^{-4} \text{T}$	$6.28 \times 10^{-4} \text{T}$	(B)	1T	$6.28 \times 10^{-4} \text{T}$	(C)	$6.28 \times 10^{-4} \text{T}$	1T	(D)	1T	1T
	If the medium is air	If the medium is iron																
(A)	$6.28 \times 10^{-4} \text{T}$	$6.28 \times 10^{-4} \text{T}$																
(B)	1T	$6.28 \times 10^{-4} \text{T}$																
(C)	$6.28 \times 10^{-4} \text{T}$	1T																
(D)	1T	1T																

Unit 9

12	A solenoid is carrying electric current. As an iron core is inserted into the solenoid, the magnetic field density of the electromagnet formed (تجربیی) a) decreases b) increases c) remains the same d) zero	
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4. True or False

No.	Question	Answer
1	A solenoid is considered as a long straight coil of wire can be used to generate a magnetic field similar to that of a bar magnet.	
2	The lines of magnetic flux through the middle of the solenoid (In a solenoid) are straight and parallel to the axis, so the magnetic field is uniform.	
3	The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it because soft iron has high permeability, so the magnetic flux lines will be concentrated.	
4	The lines of magnetic flux through the middle of the solenoid (In a solenoid) are not straight and perpendicular to the axis, so the magnetic field is uniform.	

5. Answer the following question

No.	Question	Answer
1	A certain solenoid consists of 100 turns of wire and has a length of 10 cm. Find the magnitude of the magnetic field inside the solenoid when it carries a current of 0.5 A.	
2	A certain solenoid consists of 150 turns of wire and has a length of .03 m. Find the magnitude of the Current if magnetic field inside the solenoid is 6.5×10^{-4} T	
3	Find the magnetic flux density at a point along the axis of the solenoid of length 40cm if the number of coils is 400 and current passing through on it is 5 A.	
4	Calculate the number of turns per unit length of a solenoid that carries an electric current of intensity 10A such that the magnetic flux density resulting at an interior point along its axis equals 0.05 Tesla	

Unit 9

5	what is the shape of the magnetic field due to a current in a solenoid	
6	A solenoid of length 20 cm and number of turns 200 turns, carries electric current 0.5 A, Find the magnetic flux density at the middle of the axis if: (1) The medium is air (2) An iron core is inserted inside the coil knowing that $\mu_{\text{iron}} = 2 \times 10^{-3} \text{ T.m/A}$	
7	What is the current intensity that passing in solenoid 0.5 m long and of 1000 turns produces a magnetic flux density of 0.04 Tesla at an interior point along its axis.	
8	A long solenoid has 300 turns what current is required to produce a flux density $1.2 \times 10^{-3} \text{ T}$ in the center of the solenoid if its length is 0.22 m? And what is the value of the total magnetic flux in the interior of the solenoid if its cross-sectional area is $25 \times 10^{-4} \text{ m}^2$?	
9	Calculate the number of turns per unit length of a solenoid that carries an electric current of intensity $(100/\pi)$ ampere such that the magnetic flux density resulting at the interior point along its axis equals 0.05 T, if the length is 0.6 m. Calculate its number of turns	

6. Compare between each of the following

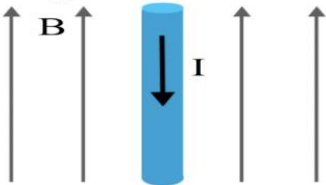
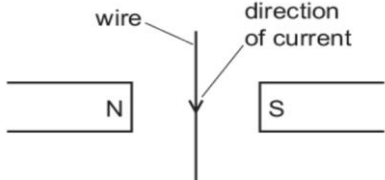
No.	Question			Answer
1	The physical relation used in calculating the value of the magnetic flux density at the center of a circular loop and at a point on the axis of a solenoid which the current traverse in each (الدور الثاني 2023)			
	Point of comparison	The magnetic flux density at the center of a circular loop	At point on the axis of a solenoid	
	The physical relation used			

Lesson (5): The magnetic force acting on a wire

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	The thumb, first finger and second finger of the left hand are all perpendicular to each other, the thumb points in the direction of motion of the wire, the pointer (first finger) points in the direction of the field, the middle (second finger) points in the direction of the current through the wire.	

2. Put circle around the correct answer

No.	Question	Choices	Answer
1	If a wire-carrying current is placed in a magnetic field such that direction of current is parallel to direction of the magnetic field a force	a) that is of maximum value acts on the wire b) that is of zero value acts on the wire c) will push the wire away from the wire d) will pull the wire towards the wire	
2	 A current-carrying wire is placed in a magnetic field B as shown below. What is the direction of the magnetic force, if any, on the wire?	a) Downward b) Into the page c) To the left d) There is no magnetic force on the wire	
3	 A current-carrying wire lies between the poles of two magnets, as shown. What is the direction of the force on the wire?	a) Into the plane of the paper b) Out of the plane of the paper c) Towards the N-pole d) Towards the S-pole	
4	The wire is placed parallel to the magnetic field so the angle θ equals	a) 0 b) 45 c) 90 d) 270	

Unit 9

5	The wire is placed parallel to the magnetic field Therefore the wire does not move $\theta = \dots$	a) 0 b) 180 c) 360 d) All of the above	
6	A length of straight conductor, carrying a current of 3A perpendicular to a magnetic field of 0.5 T, experiences a force of 1.5 N. What is the length of the conductor? (الدور الأول 2023)	a) 0.5 m b) 1 m c) 2 m d) 4 m	
8	A straight wire 35 cm long through which passes a current 10A. The wire is placed perpendicular to a magnetic field of magnetic flux density 0.5 T. Then the force acting on the wire (a) 1.75 N (c) 0.175 N	(b) 175 N (d) 1750 N	
9	the force on a wire carrying current in a uniform magnetic field is the maximum when (تجريبى) a) the current is parallel to the field lines b) the current is a 60° angle W.R.T the field lines c) the current at a 30° angle W.R.T the field line d) the current is perpendicular to the field lines		

3. Correct the underlined in the following statement

No.	Question	Answer
1	The direction of the force acting on a wire carrying-current placed in a magnetic field is identified using <u>Ampere's left-hand rule.</u>	

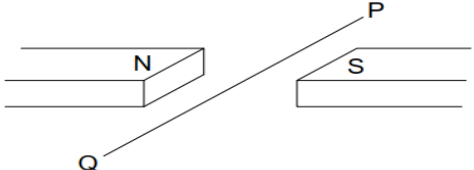
4. True or False

No.	Question	Answer
1	Amper right hand is used to determine the direction of the force (motion) of a straight wire carrying current.	
2	Magnetic force is inversely proportional with the angle between the wire and the magnetic field	

5. Answer the following question

No.	Question	Answer
1	A rectangular loop (17×10^{-4}) m^2 of 35 turns, carrying current of 2A, is placed in magnetic field of 0.35T flux density such that the plane of the loop is parallel the field the magnetic dipole moment equal A. m^2	

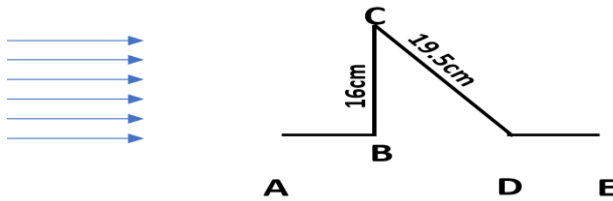
Unit 9

2	Determine the magnetic force exerts on a wire of 0.5 m length carrying a current of intensity 5 A and placed perpendicularly to the magnetic field with density 8×10^{-4} T	
3	If the magnetic force of 1 N exerts on a wire of 10 cm length carrying a current of intensity 9 A and placed perpendicularly to the magnetic field. Find the magnetic density of the field	
4	A length of 0.25 m of the wire is subject to a magnetic field strength of 40 mT. A downward magnetic force of 60 mN acts on the wire. What is the magnitude of the current? Draw the direction of the current on the picture 	
5	A straight wire 35 cm long through which passes a current 10A. The wire is placed perpendicular to a magnetic field of magnetic flux density 0.5 T. Then the force acting on the wire	
6	Calculate the magnetic force acting on a straight wire of length 50cm carrying electric current if intensity 2A placed perpendicular to magnetic flux density 0.2T.	
7	A straight wire of length 30cm carrying electric current of intensity 0.4A is placed perpendicular to the direction of a magnetic field so it is acted upon by a force of 3×10^{-4} N. Calculate the magnetic flux density	

Unit 9

	then the acting force on the wire in the same field when the angle between them is 30 degree.	
8	A straight wire of length 50cm carries a current of 1A, how can you place this wire in a magnetic field of 0.2 T in order to cause a force of 0.05 N?	
9	A straight wire of length 25cm carrying electric current of intensity 10A is placed in a uniform magnetic field of flux density 3×10^{-2} T Calculate the magnetic force acting on the wire if it: (a) is perpendicular to the magnetic field lines. (b) Makes an angle 60 degrees with the magnetic lines	
10	A straight wire of length 10cm carrying electric current of intensity 5A is placed in a uniform magnetic field of flux density 1T. Calculate the magnetic force acting on the wire when: (a) The wire is at right angle to the magnetic field. (b) The angle between the wire and the field is 45 degrees. (c) The wire is parallel to the magnetic flux lines.	
11	Determine the magnitude and direction of the force acting on horizontal conductor of 5cm long and carrying 30Ampere, flowing from south to north, if it is placed in air in a horizontal magnetic flux of density 0.8T is directed from east to west.	
12	Find the force on each segment of the wire shown in the figure, knowing $B=0.15$ T.	

Unit 9

	Assume the current in the wire to be 5A. AB and DE lengths are 5cm. 	
13	The resulted magnetic force affected on wire carrying an electric field putted in magnetic field is vanished when the wire is be (1) Perpendicular to the field (2) parallel to the field (3) Inclined with the field by angle 30 degrees (4) Inclined with the field by angle 60 degrees	

6. Compare between each of the following

No.	Question			Answer
1	Ampere's right-hand rule & Fleming's left-hand rule in terms of the use of each (الدور الأول 2024) (الدور الثاني 2023)			
	Point of comparison	Ampere's right-hand rule	Fleming's left-hand rule	
	The use			

Lesson (6): The magnetic torque acting on a coil

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	It is the magnetic torque acting on a coil carrying a current placed parallel to a uniform magnetic field of flux density of 1 tesla (الدور الأول 2024)	
2	It is a vector emanating from North Pole of the coil and perpendicular to its area	

2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field. (الدور الأول 2023)	
2	magnetic torque direction	
3	magnetic dipole moment	

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	The magnetic torque acting on a loop carrying-current placed in a magnetic field is maximum when	a) the plan of the coil is perpendicular to the magnetic field lines b) the plan of the coil is at an angle of 30° with the magnetic field lines c) the plan of the coil is at an angle of 30° with the magnetic field lines d) the plan of the coil is parallel to the magnetic field lines	
2	A current carrying rectangular coil of wire is placed in a magnetic field the magnitude of the torque on the coil is not dependent upon which one of the following quantities	a) the magnitude of the current in the loop b) the area of the loop c) the direction of the current in the loop	

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		d) the orientation of the loop	
3	The magnetic torque measured in	a) Ampere b) N c) Weber d) N.m	
4	It is the magnetic torque acting on a coil carrying current placed parallel to a uniform magnetic field of flux density of 1 tesla	a) The dipole moment b) Tesla c) Force d) None	
5	The torque acting on a coil carrying current in a uniform magnetic field becomes zero when the plane of the coil is to the direction of the magnetic field. (الدور الثاني 2023)	a) perpendicular b) parallel c) inclined by an angle 30° d) inclined by an angle 60°	

4. Correct the underlined in the following statement

No.	Question	Answer
1	The concept of the magnetic torque acting on the coil carrying current is the idea on which is based many instruments as the galvanometer and the electric <u>generators</u> .	
2	The magnetic torque acting on the coil carrying current is directly proportional to <u>electric</u> field density.	
3	The magnetic dipole moment $ md $ is given by the following relation $ md = IAN\sin\theta$ (الدور الأول 2023)	

5. True or False

No.	Question	Answer
1	When the perpendicular to coil's plane perpendicular to the magnetic field $\tau = 0$	
2	When the perpendicular to coil's plane perpendicular to the magnetic field the coil's plane is parallel to the magnetic field	

6. Answer the following question

No.	Question	Answer
1	A rectangular coil of 500 turns has dimensions $0.2\text{m} \times 0.1\text{m}$, carries a current 5A. The plane of the coil makes angle 60° to a uniform magnetic field of flux density 0.8 T. Calculate the magnetic torque acting on the coil	

Unit 9

2	A rectangular coil has dimensions $0.15\text{m} \times 0.12\text{m}$, carries a current 5A . The plane of the coil makes angle 45° to a uniform magnetic field of flux density 0.65 T if the magnetic torque acting on the coil is 40 N.m . Calculate number of turns.	
3	A rectangular loop of current-carrying wire between the poles of a magnet that produces a field with a magnitude of 250 mT . The loop has a magnetic dipole moment of $500\text{ }\mu\text{N}\cdot\text{m/T}$	
4	Mention the application of magnetic torque	
5	A circular coil has 100 turns and carries a current of 10 A , it has a cross sectional area of 0.3m^2 and is placed in magnetic field of flux density 0.2T . Calculate the maximum value of the torque acting on the coil and determine the position of the coil with respect to the field in this case	
6	a rectangular coil of 100 turns and dimensions $(5 \times 2\text{cm})$ carrying a current of 2A is suspended vertically in a uniform magnetic field of 0.2T acting normal to the plane of the coil. Find the torque acting on the coil.	
7	a loop of cross-sectional area 0.2m^2 and 500 turns, current of 10 A is placed at 30° between the perpendicular to its plane and a magnetic field of 0.25T flux density. Calculate the torque acting on loop	
8	A loop of 200 turns current of 10 A and cross-sectional area 0.2m^2 is placed in magnetic field of 0.4T , Calculate the torque acting on the loop when the plane of the loop and the magnetic field is 60° .	
9	A circular coil of 100 turns and diameter 0.2 m carrying current of 5A is suspended vertically in horizontal magnetic field of 1.2T . The coil rotates by an angle 30° with the normal to the magnetic field. Find the torque acting on the coil	
10	A circular coil of 50 turns has radius 20cm carries a current 3A . the plane of the coil	

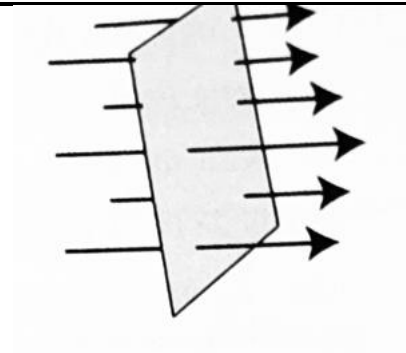
Unit 9

	makes angle 60° to magnetic field whose flux density uniform of 0.5 T. Calculate the magnetic torque acting the coil. (تجريبى)	
11	A rectangular coil of 500 turns has dimensions $0.2\text{m} \times 0.1\text{m}$, carries a current 5A. The plane of the coil makes angle 60° to a uniform magnetic field of flux density 0.8 T. Calculate the magnetic torque acting on the coil. (الدور الأول 2023)	
12	A loop of cross-sectional area 0.1m^2 and 100 turns, carries a current of intensity 20 A is placed parallel to a uniform magnetic field of flux density 0.5 T. Calculate the torque acting on the loop. (الدور الثانى 2023)	

7. Compare between each of the following

No.	Question		Answer
1	Mention the scientific base of the following.		
	Point of comparison	galvanometer	electric motor
	scientific base		
2	Compare		
	Point of comparison	Magnetic dipole moment	Magnetic permeability
	Measuring unit		

Unit 9

			
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